Leave Balance Converter

Technical Overview & Migration Handover

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1. The Problem We Are Solving

Clients currently managing leave balances within Simplifi without an API connection face a painful three-step manual process:

- Export a leave balance report from their payroll system (Smartly, IMS/MYOB, Rocket or Crystal).
- Manually reformat and remap the data — renaming columns, converting units, removing redundant rows — in Excel or a text editor.
- Re-upload the transformed file into Simplifi.

This is error-prone, time-consuming, and entirely avoidable. Each manual transformation step introduces the risk of incorrect mappings, wrong unit conversions (hours vs days), or stale data being imported.

> *On the question of API connectivity: Smartly and IMS/MYOB do not currently expose public APIs to Simplifi for leave balance data. While a direct integration would be the ideal long-term solution, it is simply not available today. In the meantime, expecting clients to absorb a more manual or degraded workflow is not a realistic option — particularly when competing platforms already offer seamless leave balance imports. This tool bridges that gap now, with no burden placed on the client.*

2. What the Tool Does

The Leave Balance Converter is a lightweight, single-page web application that eliminates the manual transformation step entirely. A payroll administrator:

- Selects their payroll provider (Smartly, IMS/MYOB, Rocket or Crystal).
- Drops or browses to their exported leave balance file (CSV, Excel or PDF).
- Chooses optional output settings (skip zero balances, select specific leave codes).

- Reviews a live preview table and downloads a Simplifi-ready CSV in one click.

The entire workflow takes under 60 seconds and requires no technical knowledge.

3. Technical Architecture

The application is intentionally simple by design — a deliberate architectural choice to maximise security, minimise maintenance overhead, and eliminate infrastructure dependencies.

Type	Static single-page application (SPA)
Language	Vanilla HTML5, CSS3, and JavaScript — no framework, no build step
File count	One file: index.html (all logic, styles, and markup are self-contained)
Dependencies	No npm packages. Loaded from cdnjs only when needed: SheetJS 0.18.5 for Excel files (Crystal) and pdf.js 3.11.174 for PDFs (IMS Total Leave).
Data processing	All parsing (CSV, Excel, PDF) and transformation runs in the user's browser via the Web APIs (FileReader, Blob, URL.createObjectURL)
Server	None — the application has no backend, no API calls, and no server-side logic
Browser support	All modern browsers (Chrome, Edge, Firefox, Safari)

Provider support is implemented as a plug-in pattern inside the JavaScript. Each provider has:

- A detect() function that reads the first few lines of the uploaded file to identify the format automatically.
- A parse() function that reads the provider-specific CSV structure into a normalised row array.
- A convert() function that maps the normalised rows into Simplifi's import column format.

Current providers: Smartly, IMS/MYOB, Rocket and Crystal. IMS/MYOB has three formats (Employee Leave CSV, Total Leave CSV and Total Leave PDF), all under the IMS / MYOB button and told apart automatically.

4. Security & Data Privacy

Data privacy was a primary design constraint. The following guarantees are built into the architecture:

> *No employee data — names, IDs, balances, or any other personal information — ever leaves the user's device. The application does not transmit, store, log, or cache any data from uploaded files.*

Network requests	No analytics or telemetry, and no file data is ever sent. The only later requests are script downloads of SheetJS or pdf.js from cdnjs, made when an Excel or PDF file is dropped.
PII ingestion	None. File contents are read by the browser's FileReader API and processed entirely in JavaScript memory. Data is never written to disk, a database, or any remote endpoint.
Data retention	None. All processed data lives in JavaScript variables and is discarded when the tab is closed or the user clicks "Start over".
Authentication	Not required. The tool is intentionally public-facing within the Simplifi ecosystem — access control is handled at the hosting layer.
File upload	No upload occurs. The browser reads the file locally via the File API. The "upload" is a local file-read operation only.
Dependencies	SheetJS and pdf.js, loaded on demand from cdnjs at pinned versions. They run locally and send no data. To remove the CDN dependency, host copies alongside index.html.

The footer of the application displays the following notice, visible to all users:

> *"All data is processed locally in your browser — nothing is sent to any server."*

5. Current Deployment

The tool is currently hosted via GitHub Pages from a personal GitHub repository. Details:

Repository	https://github.com/Jamie-Callawick/Leave-Balance-Worker-App

Owner	Personal account — Jamie Callawick (Jamie-Callawick)
Branches	main (production), Dev (development / staging)
Hosting	GitHub Pages — served as a static site directly from the repository
URL format	https://jamie-callawick.github.io/Leave-Balance-Worker-App/ (current)
Build process	None — changes pushed to main are immediately live. No CI/CD pipeline is currently configured.
Access	Publicly accessible via the GitHub Pages URL

6. Migration: Personal GitHub to Simplifi Organisation

Moving the repository from a personal account to the Simplifi GitHub organisation is a straightforward process. The steps below assume the Simplifi org already exists on GitHub. No code changes are required.

1	**Create org repository**	Create a new repository called Leave-Balance-Worker-App (or equivalent) inside the Simplifi GitHub organisation. Set visibility to match your policy (private recommended during transition).
2	**Transfer or mirror**	Option A (recommended): Settings > Danger Zone > Transfer ownership > select the Simplifi org. Preserves all git history, branches, and issues. Option B: Clone and push to a new org repo — use only if a clean slate is preferred.
3	**Update GitHub Pages**	In the new org repository, go to Settings > Pages > set Source to the main branch (root). The public URL will become https://simplifi.github.io/Leave-Balance-Worker-App/ (adjust for org name).
4	**Update any shared links**	If the old GitHub Pages URL has been shared with clients or in documentation, update references to the new URL. GitHub Pages does not redirect the old URL after a transfer.
5	**Branch protection**	Apply branch protection rules to main: require pull request reviews before merge, disable force-push. This prevents accidental direct pushes to production.
6	**Team access**	Add relevant Simplifi dev team members with appropriate roles (Write or Maintain). Remove Jamie's personal admin access if the account will no longer manage the repo.
7	**Custom domain (optional)**	To use a branded URL (e.g. tools.simplifi.work/leave-converter), add a CNAME file to the repo and configure the DNS record to point to github.io per GitHub Pages documentation.

8	**Verify & smoke test**	After migration, load the new URL, upload a sample file for each provider (CSV, .xls, PDF) and confirm the output is correct. Check browser console for any errors.

7. Risks & Mitigations

The following risks have been identified across hosting, security, and ongoing maintenance. Each is rated by likelihood and impact.

Risk	Likelihood	Impact	Mitigation
GitHub Pages outage	Low	Medium	GitHub Pages has a strong uptime record. If availability is a concern, the single index.html file can be mirrored to any static host (Netlify, Cloudflare Pages, Azure Static Web Apps) with minimal effort.
Personal account access lost	Low	High	The repository sits under a personal GitHub account. If access is lost before migration, the codebase must be re-created from a local copy. Mitigate by completing the migration to the Simplifi org promptly and keeping a local backup of index.html.
No uptime SLA	Medium	Low	GitHub Pages does not offer an SLA. For an internal-use tool this is acceptable. If an SLA becomes a requirement, move hosting to a managed platform such as Azure Static Web Apps.
Tool URL changes post-migration	High	Medium	After migrating to the Simplifi org the GitHub Pages URL will change, breaking any existing bookmarks or documentation links. Communicate the new URL to all users before cutover and update all internal documentation.
Scope creep into PII handling	Low	High	The current tool handles only leave balance figures. If future requirements add employee names or IDs to the output, a privacy review is required before release to ensure the no-PII guarantee remains valid.
No dedicated maintainer post-handover	Medium	Medium	If the dev team does not assign an owner, bug reports or provider format changes (e.g., Smartly updates their CSV export) will go unaddressed. Assign a named maintainer and add the repo to standard team monitoring.

Risk	Likelihood	Impact	Mitigation
Publicly accessible URL	Medium	Low	The tool is accessible to anyone with the link. No sensitive data is processed, so this is low risk. If policy requires restricted access, Cloudflare Access or a GitHub org-only Pages restriction can be applied without code changes.
No automated tests	Medium	Medium	There are no unit or integration tests. A provider format change could silently break conversions. Add a GitHub Actions workflow that runs sample CSV conversions and validates the output on every pull request.

8. Future Considerations

The following enhancements are suggested as the tool matures and API availability changes:

- Other payroll support — there is a placeholder provider slot already stubbed in the code; only the detect/parse/convert functions need to be implemented once a sample export is available.
- Direct API integration — if Smartly or IMS/MYOB release API access to leave balance data in the future, the transformation logic in this tool can be reused server-side to enable a true one-click sync.
- CI/CD pipeline — a simple GitHub Actions workflow can lint, validate, and optionally run smoke tests on pull requests before merge.
- Version pinning — embedding a version number in the UI footer (e.g., v1.2.0) will help support staff identify which version a client is running when troubleshooting.
- Access control — if the tool needs to be restricted to Simplifi clients only, authentication can be added at the CDN/hosting layer (e.g., Cloudflare Access) without changing any application code.

9. Change Log

Changes made to the converter since this document was first prepared, newest first. Add a new entry for each release.

Date	12 November 2026
Provider	IMS/MYOB
Change	Added support for the IMS/MYOB TOTAL LEAVE REPORT printed to PDF (e.g. "TOTAL LEAVE REPORT.pdf"). This is the same report as the Total Leave CSV, in its printed layout.
Reason	Requested so the IMS Total Leave report can be converted when it is supplied as a PDF rather than a CSV. Previously the tool rejected PDFs.
Conversion logic	New provider "ims-total-leave-pdf", which shares the IMS / MYOB button. pdf.js 3.11.174 is loaded from cdnjs only when a .pdf is dropped. The text is rebuilt into lines from each text item's on-page position, so page rotation is handled. The PDF is recognised from its titles (TOTAL LEAVE REPORT, ANNUAL LEAVE, SICK LEAVE). Each employee line is read as: employee number, name, AL due date, outstanding, accrued, advanced and balance (hrs), SL due date and balance (days), and Alt balance (days). Blank AL or SL sections, e.g. new starters, are allowed. Rows use the same field names as the Total Leave CSV, so convert() is shared with "ims-total-leave". Employees with no employee number are skipped. Scanned PDFs, or PDFs of any other report, are rejected with a message.
On-screen indicator	The status line reads "IMS / MYOB (Total Leave report — PDF) format detected", followed by the company name and reporting date. The leave code grid shows six columns with their units, all on by default. The notice above the preview lists each column's unit and names any employees skipped for having no number. It also shows a GRAND TOTALS cross-check: the tool sums every balance it read and compares it to the report's GRAND TOTALS line. A match shows a tick; a mismatch shows a warning to check the file or use the CSV export.

CSV output	Columns: Employee Code, AL Outstanding, AL Accrued, AL Advanced, AL Balance, SL Balance and Alt Balance. Anniversary date columns follow each balance when that option is on (AL from AL Due Date, SL from SL Due Date). AL figures are in HOURS; SL Balance and Alt Balance are in DAYS. Values have 2 decimal places. Unlike the Total Leave CSV, the PDF includes AL Accrued and AL Balance (Outstanding + Accrued − Advanced). Choose which AL figure(s) to import.
Backwards compatibility	CSV and Excel handling is unchanged. The Total Leave CSV, Employee Leave, Smartly, Rocket and Crystal produce identical output.
Tested with	TOTAL LEAVE REPORT.pdf (Holmwood Lifecare Limited, reporting date 11/10/2026): 72 employees converted, and all six columns match GRAND TOTALS. For example, employee 102304 has AL Outstanding 142.24, Accrued 19.69 and Balance 161.93 hrs, SL 4 days and Alt 1 day. One employee (MCLENNAN, Julie) has no employee number and is skipped with a warning. Regression checked against Total_Leave_2026_09_27 (CSV): 72 employees, output unchanged.
Rollback	The previous version is git commit c7de31f in the repository.

Date	2 October 2026
Provider	IMS/MYOB
Change	Added support for a second IMS/MYOB export format, the "Total Leave" report (e.g. Total_Leave_2026_09_27.csv).
Reason	The Total Leave file was not recognised as IMS. It fell through to the Smartly parser, so the converted output was unusable.
Conversion logic	New provider "ims-total-leave", which shares the IMS / MYOB button. detect(): a single header row containing Employee Number, AL Outstanding and SL Balance. parse(): one row per employee; rows with no Employee Number are skipped. convert(): Employee Code = Employee Number, and AL Outstanding, AL Advanced, SL Balance and Alt Balance are passed through as-is. Dollar-value columns are not output.
On-screen indicator	The status line reads "IMS / MYOB (Total Leave report) format detected". The leave code grid shows the four columns with their units (all on by default). A blue notice above the preview lists the unit of each column, read from the file's AL Unit, SL Unit and Alt Unit columns.

CSV output	Columns: Employee Code, AL Outstanding, AL Advanced, SL Balance, Alt Balance. Units are mixed: AL Outstanding and AL Advanced are in HOURS, SL Balance and Alt Balance are in DAYS. Map each accordingly in the import. This report has no AL Accrued figure; use the Employee Leave report if accrued is needed.
Backwards compatibility	The original IMS/MYOB Employee Leave format, Smartly and Rocket are unaffected and produce identical output.
Tested with	Total_Leave_2026_09_27 (IMS): 72 employees converted. For example, employee 102304 has AL Outstanding 150.24 hrs, SL Balance 5 days and Alt Balance 1 day. One row (no Employee Number, all-zero balances) is skipped. Regression checked against Eldon IMS Employee Leave (85), Rocket VIEWTTLB (186) and Mckenzie HC Smartly (524 records).
Rollback	Previous version is git commit 6ec97c5 in the repository.

Date	28 September 2026
Provider	Smartly
Change	New optional "AL Accrued" output column carrying the Annual Leave accrued value.
Reason	Requested so the Annual Leave accrued figure from Smartly comes through to Simplifi alongside the #70 balance.
Conversion logic	On Annual Leave rows, the "Accrued" column is passed through as-is (no unit conversion) into "AL Accrued", placed after #70. It is captured separately from the balance, so with "skip zero balances" on, a zero #70 does not hide a non-zero accrual; a zero or blank accrual is left blank.
On-screen indicator	AL Accrued appears as its own tick box in the leave code grid (on by default), labelled "Annual Leave Accrued (weeks, raw from Smartly)", and as a column in the preview.
CSV output	Adds an "AL Accrued" column after #70. Values are in WEEKS (the Unit Smartly reports for Annual Leave), not hours — map it as weeks in the import, or untick it if not needed.
Backwards compatibility	All existing columns and values are unchanged. Unticking AL Accrued gives output identical to the previous version. IMS/MYOB and Rocket are unaffected.
Tested with	Mckenzie HC (PPE 12/04/2026): 95 employees, e.g. employee 351 #70 = 69.716 hrs, AL Accrued = 3.7145 weeks. Checked with skip zero balances on and off.
Rollback	Previous version saved as index.backup-2026-09-28-pre-accrued.html alongside index.html.

Date	28 September 2026
Provider	Smartly
Change	Annual Leave (#70) now falls back to Balance (Days) when Balance (Hours) is blank.
Reason	Some Smartly exports (first seen with Age Care Central Ltd) include an extra Balance (Days) column. On Annual Leave rows Balance (Hours) is blank and the figure is only in Balance (Days). Before this change #70 was output blank for every employee in these files.
Conversion logic	Annual Leave: use Balance (Hours) if populated, otherwise Balance (Days). All other leave types continue to use Balance. Columns are matched by header name, so the extra column does not affect parsing.
On-screen indicator	When any #70 value came from days, the preview column header shows "#70 (Days)" and an amber notice warns that Annual Leave is in days. If a file mixes hours and days across employees, the header shows "#70 (Hours/Days)" with a mixed-units warning. Zero balances are ignored when deciding the unit.
CSV output	Unchanged: the downloaded file keeps the plain "#70" header so the Simplifi import mapping is not affected. The unit label appears in the preview only, so the person importing must treat #70 as days for these clients.
Backwards compatibility	Exports without a Balance (Days) column (e.g. Mckenzie HC) produce identical output to the previous version.
Tested with	Age Care Central Ltd (PPE 09/08/2026): 83 employees, all #70 values populated in days, notice shown. Mckenzie HC (PPE 10/05/2026): 92 employees, output unchanged, no notice.
Rollback	Previous version saved as index.backup-2026-09-28.html alongside index.html.

10. Point of Contact

For questions about the codebase, the current deployment, or the migration process, contact:

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